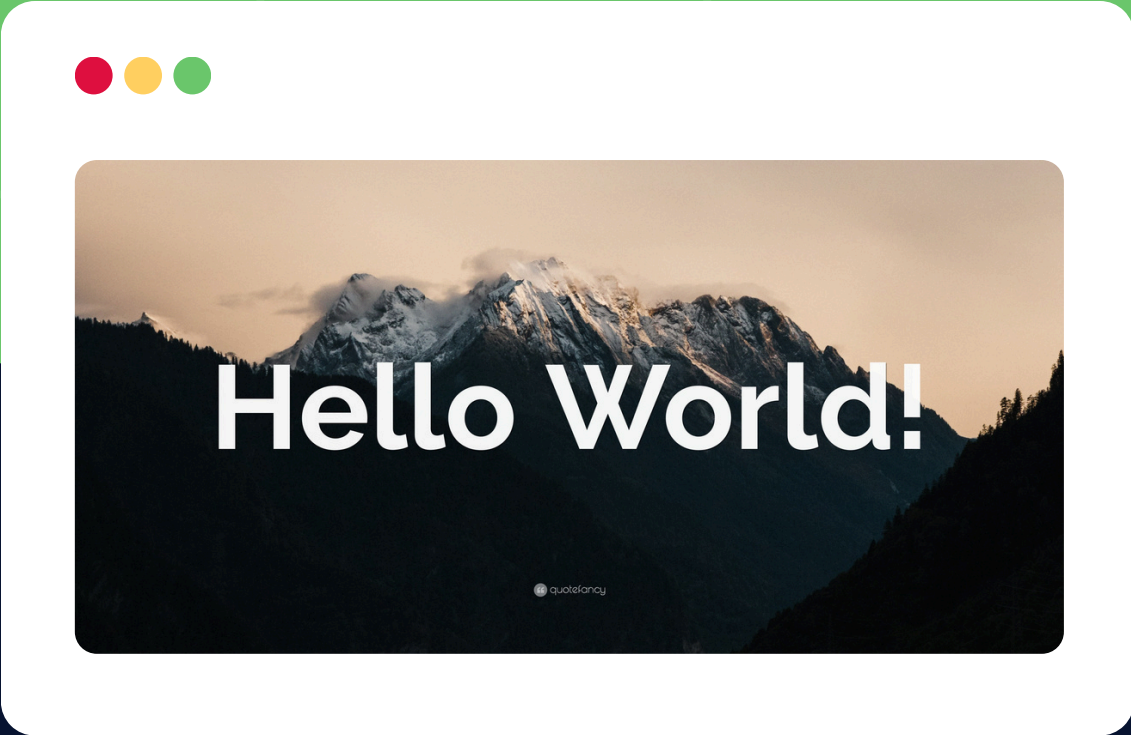
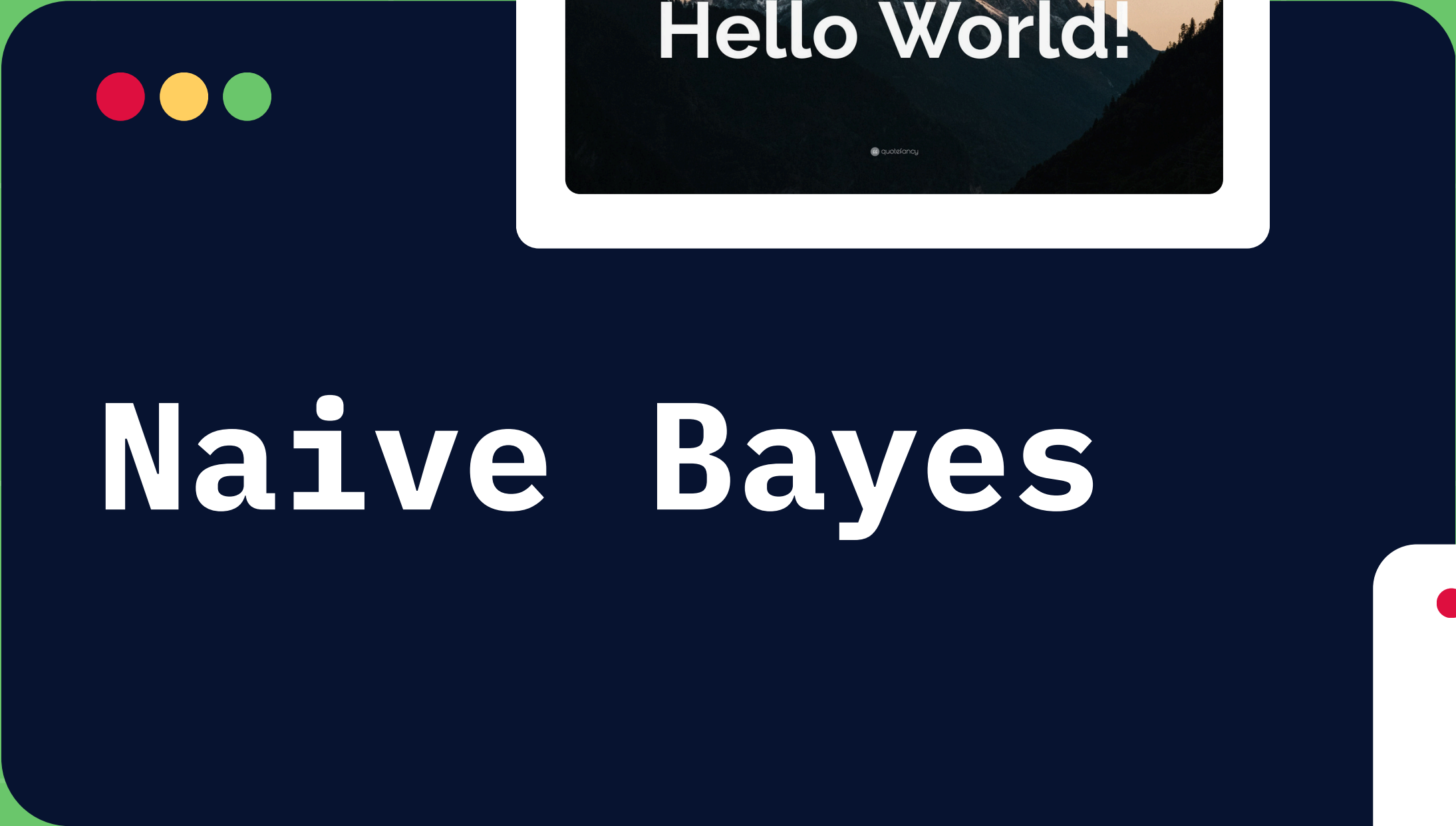
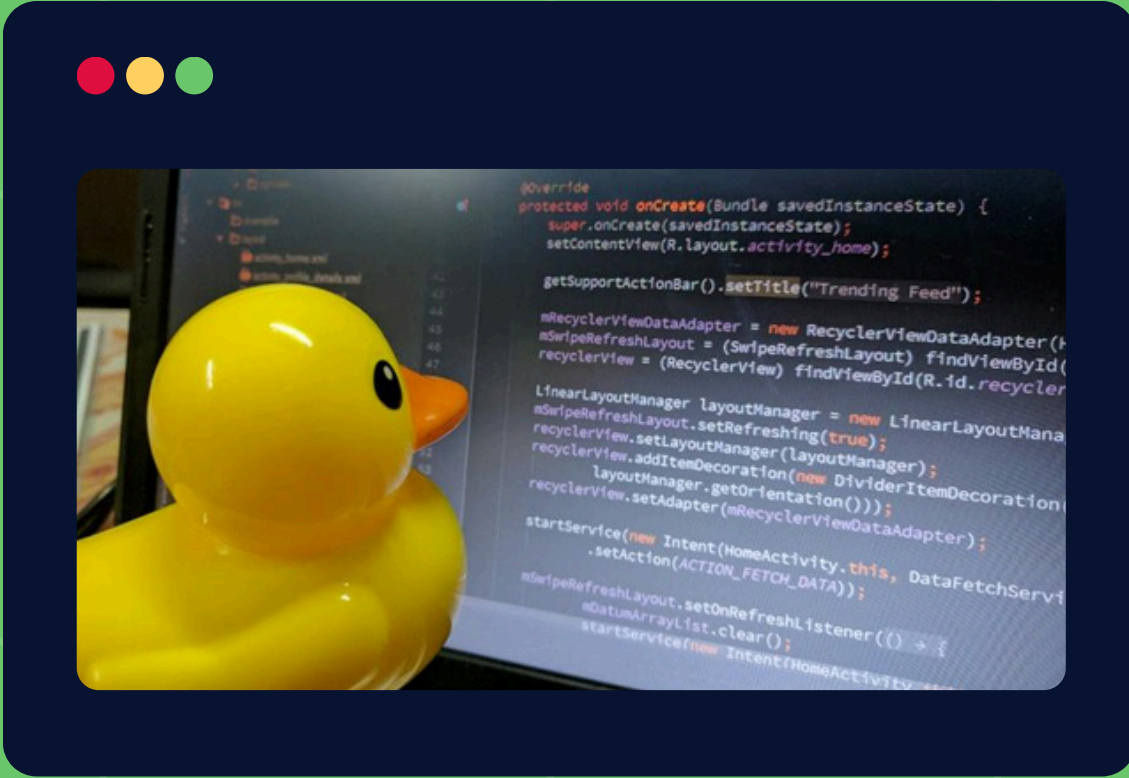




Hello World!



Naive Bayes



Session -> 6



Discriminative **vs** Generative classifiers

Introduction – Voice Classification Example

- Imagine we have a dataset of voice recordings – some from male speakers, and some from female speakers.
- Our goal is to build a system that listens to a new voice and tells us: **Is this voice male or female?**

Input: record

target: female, male

problem: classification problem

Discriminative Classifiers

- Models that directly learn the boundary between classes.

What do they learn?

→ the probability of the label (y) given feature (x)

$$P(\text{label} \mid \text{features}) = P(y \mid x)$$

Goal → Just classify

Examples: Logistic Regression, Support Vector Machine

Generative Classifiers

- Models that learn how each class is distributed, then use that to classify or generate data.

What do they learn?

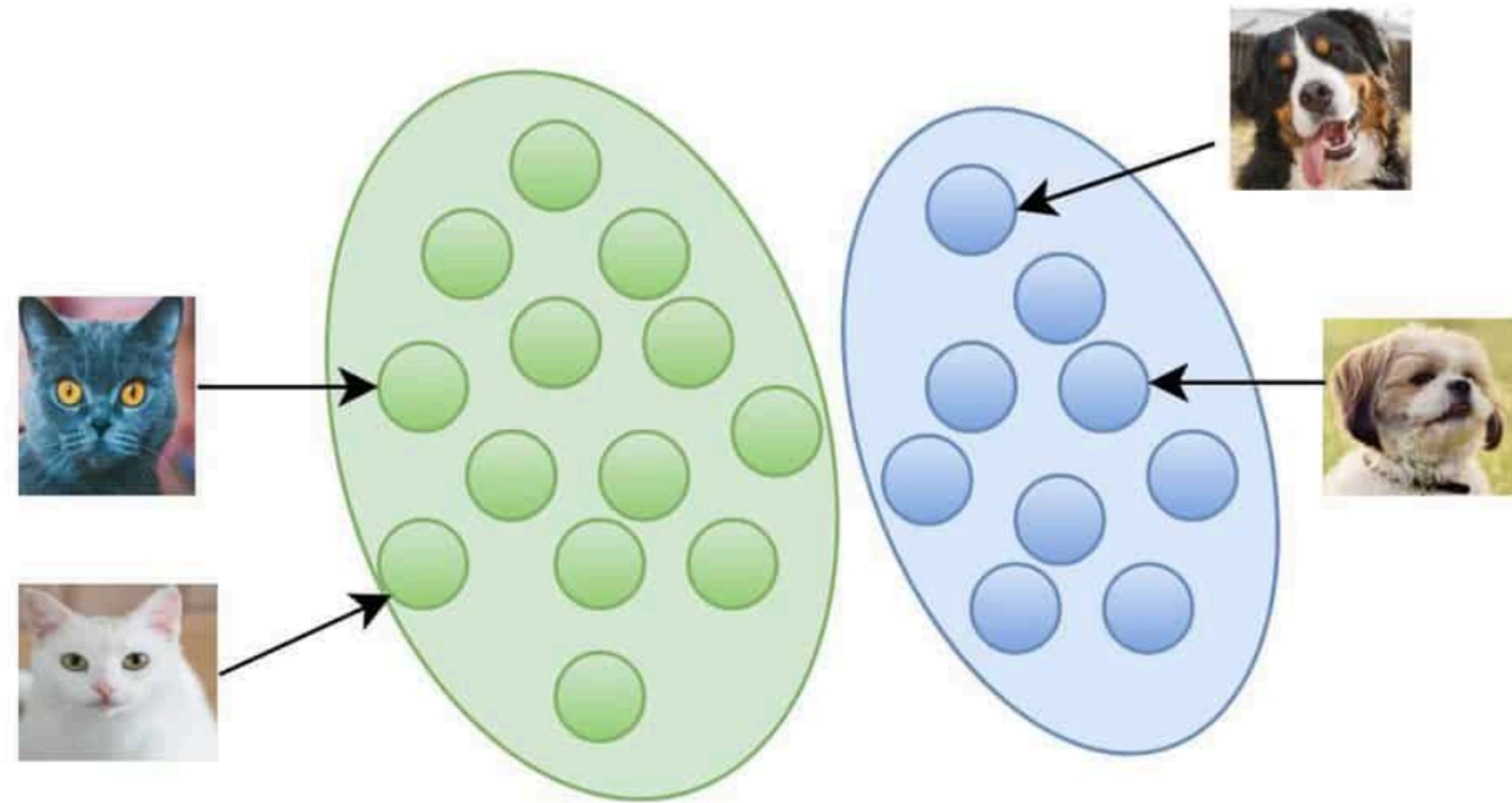
→ the probability of features (x), given the class label (y)

$$P(\text{features} \mid \text{label}) = P(x \mid y) P(y)$$

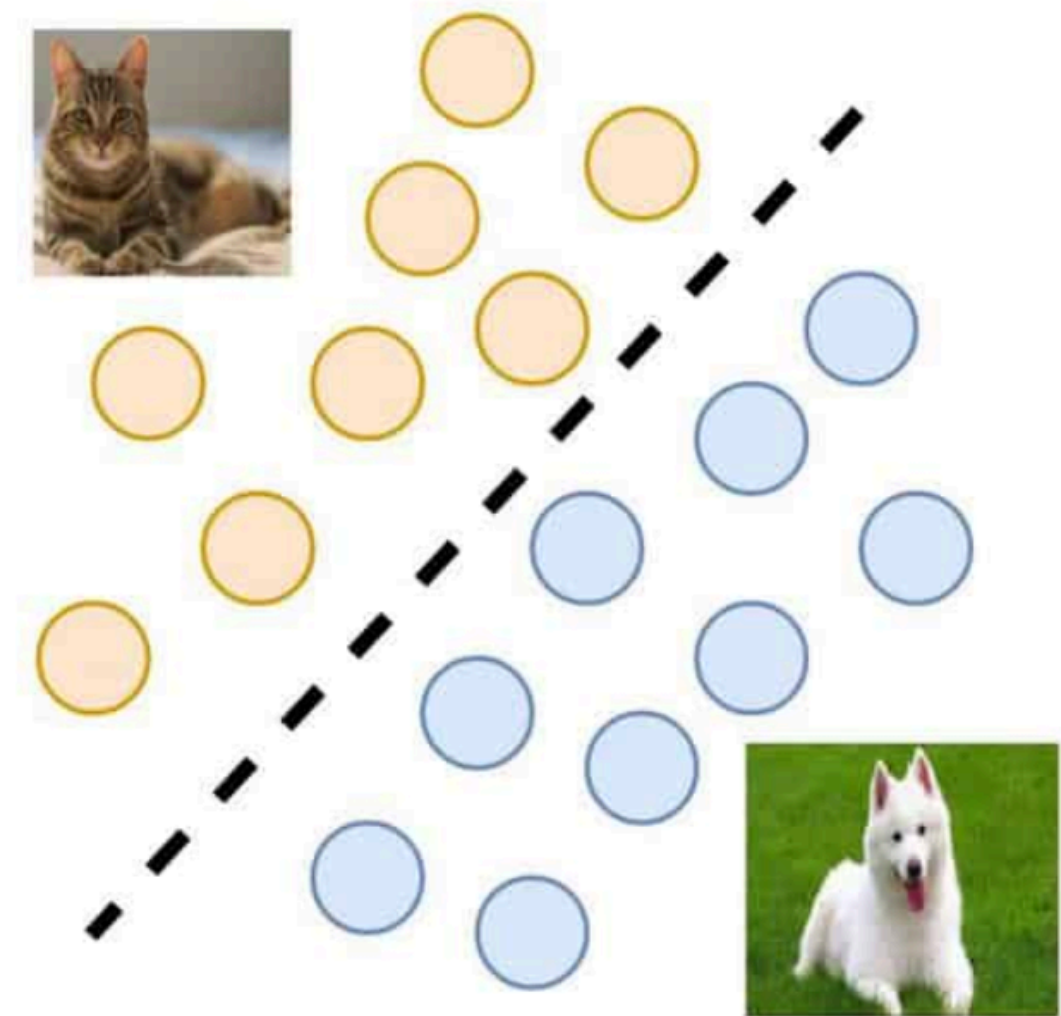
Goal → classify data, or generate new samples

Examples: Naive Bayes, GDA

Generative vs Discriminative



Generative



Discriminative



Naïve Bayes

What is Naïve Bayes

- Naïve Bayes is a probabilistic **classification algorithm** based on Bayes' Theorem.

Why is it called Naïve?

→ It assumes that features are **conditionally independent**, that knowing one feature does not tell you anything about the other features.

Bayes' Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

- **$P(A | B)$** : Probability of class A given features B (this is what we want).
- **$P(B | A)$** : Probability of features B given class A.
- **$P(A)$** : Prior probability of class A.
- **$P(B)$** : Probability of features B .

How does it work?

1- Calculate Class Probabilities

→ For each class, calculate its probability (Prior)

$$P(C_1) = \frac{\text{Number of samples in class } C_1}{\text{Total number of samples}}$$

How does it work?

2- Calculate Feature Probabilities – $P(X^i \mid C_1)$

→ This is the likelihood: How likely each word (feature) appears in class C_1 .

$$P(x^i \mid C_1) = \frac{\text{Count of word } x^i \text{ in class } C_1 + 1}{\text{Total word count in class } C_1 + V}$$

How does it work?

3- Given an Input, Compute Posterior – $P(C_1 | X)$

→ We combine all the probabilities using Bayes' Theorem:

$$\log P(C_1 | X) = \log P(C_1) + \sum_i \log P(x^i | C_1)$$



Example

Spam or Ham Classifier

- We have a dataset of some words that appeared in both spam and ham e-mails with their frequencies

Email	Label
"Win cash now"	Spam
"Win a free lunch"	Ham
"Urgent prize claim"	Spam
"Meeting about prize"	Ham
"Free lottery ticket"	Spam

Spam or Ham Classifier

- **Firstly**, let's compute the probability of an e-mail being a spam and the probability of being a ham (class prior):

$$P(\text{Spam}) : 3/5 = 0.6$$

$$P(\text{Ham}) : 2/5 = 0.4$$

Spam or Ham Classifier

- **Secondly**, compute each word frequency per class, then the probability of that word given each class

Word	count in spam	count in ham	P(Word Spam)	P(Word Ham)
win	1	1	$1/9 \approx 0.111$	$1/7 \approx 0.143$
prize	1	1	$1/9 \approx 0.111$	$1/7 \approx 0.143$
free	1	1	$1/9 \approx 0.111$	$1/7 \approx 0.143$
cash	1	0	$1/9 \approx 0.111$	0
urgent	1	0	$1/9 \approx 0.111$	0
lunch	0	1	0	$1/7 \approx 0.143$
meeting	0	1	0	$1/7 \approx 0.143$

Spam or Ham Classifier

- **At last**, classify the new e-mails(input)
input example → "Win free prize"

$$\begin{aligned} 1- P(\text{Spam} | \text{"Win free prize"}) &= \\ P(\text{Spam}) \times P(\text{"win"} | \text{Spam}) \times P(\text{"free"} | \text{Spam}) \times P(\text{"prize"} | \text{Spam}) \\ &= 0.6 \times 0.111 \times 0.111 \times 0.111 \approx 0.00082 \end{aligned}$$

$$\begin{aligned} 2- P(\text{Ham} | \text{"Win free prize"}) &= \\ P(\text{Ham}) \times P(\text{"win"} | \text{Ham}) \times P(\text{"free"} | \text{Ham}) \times P(\text{"prize"} | \text{Ham}) \\ &= 0.4 \times 0.143 \times 0.143 \times 0.143 \approx 0.00117 \end{aligned}$$

Final output: Ham (with a higher score or 0.00117)



Laplace Smoothing

What Happens When Naive Bayes Sees Something New?

- Imagine a new email contains a word not in the training set: "Click here to claim your prize!"
- The word "**click**" wasn't in the training data.
- In Naive Bayes:
 - $P(\text{"click"} | \text{Spam}) = 0$
 - $P(\text{"click"} | \text{ham}) = 0$

Since probabilities are multiplied, any zero makes the whole prediction zero.

What Happens When Naive Bayes Sees Something New?

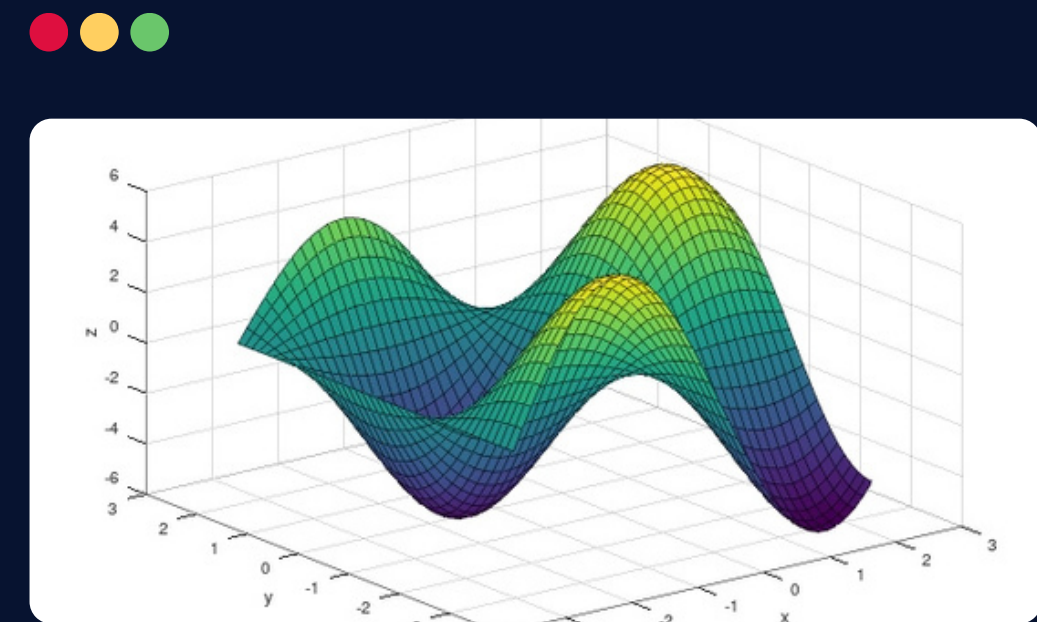
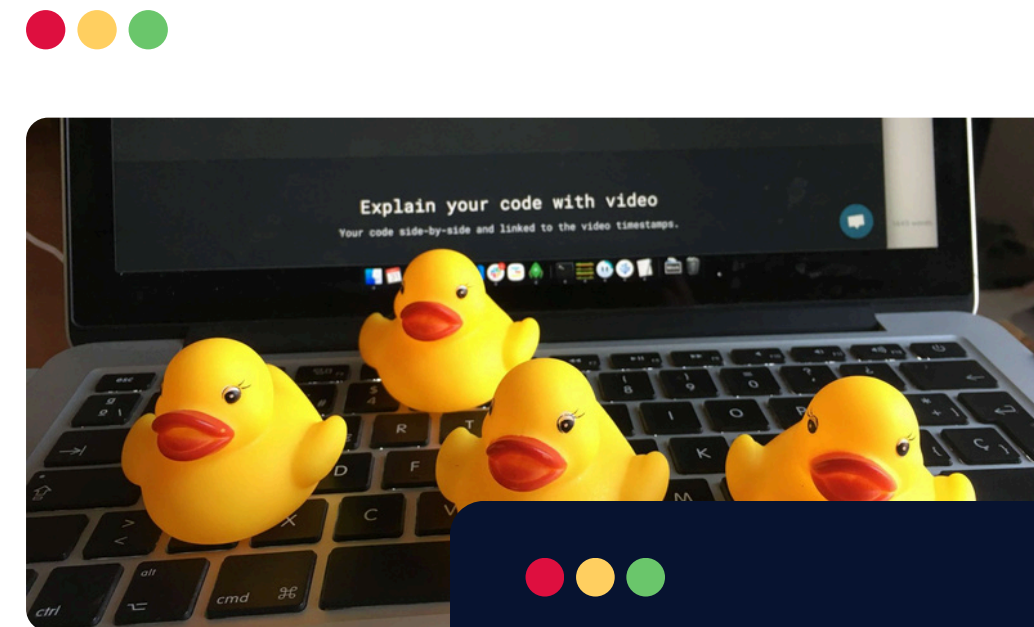
- For example:
 - $P(\text{"click"}|\text{Spam}) = 0.06 \times 0.07 \times 0 = 0$
 - $P(\text{"click"}|\text{ham}) = 0.1 \times 0.005 \times 0.0001 = 5 \times 10^{-9}$

Result: Email 1 is classified as Ham, even though most words look spammy – because of one zero!

Solution: Laplace smoothing

- Laplace smoothing solves the zero probability problem in Naive Bayes by assuming that every word appears at least once in every class.

$$P(x^i | C_1) = \frac{\text{Count of word } x^i \text{ in class } C_1 + 1}{\text{Total word count in class } C_1 + V}$$



Thank You

created by:
Ahd Ehab